Mean Flows for One-step Generative Modeling

Research focus: does interval normalization change stop-gradient training?

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Selected paper: Geng et al., arXiv:2505.13447v1, 2025.

Why did I choose MeanFlow?

Selected paper: Mean Flows for One-step Generative Modeling — Geng et al., 2025.

PRACTICAL VALUE

Reduce sampling cost; study quality versus compute.

CLEAR MATHEMATICS

An exact identity becomes a neural training target.

MY QUESTION

How does normalization affect stopped-target training?

Images generated in one evaluation



Source: MeanFlow v1, Fig. 5 (cropped).
Author-curated samples, not my results.

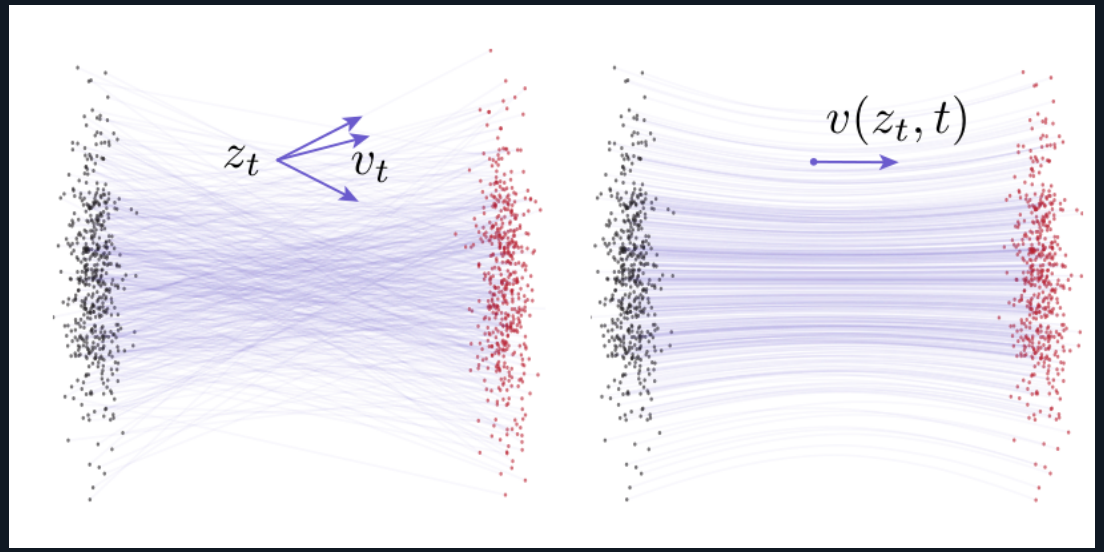
Paper evidence: ImageNet 256 × 256, 1 NFE, FID 3.43 (XL/2; pretrained VAE).

How is MeanFlow related to machine learning?

It changes what a neural network learns and how that network is optimized.

GENERATIVE LEARNING

Learn noise-to-image transport.
Flow Matching: instantaneous velocity.
MeanFlow: interval-average velocity.
Train with data, JVP and stopped targets.



Source: MeanFlow v1, Fig. 2 (cropped).
Left: conditional flows. Right: marginal flows.

My focus: representation → optimization → generative performance.

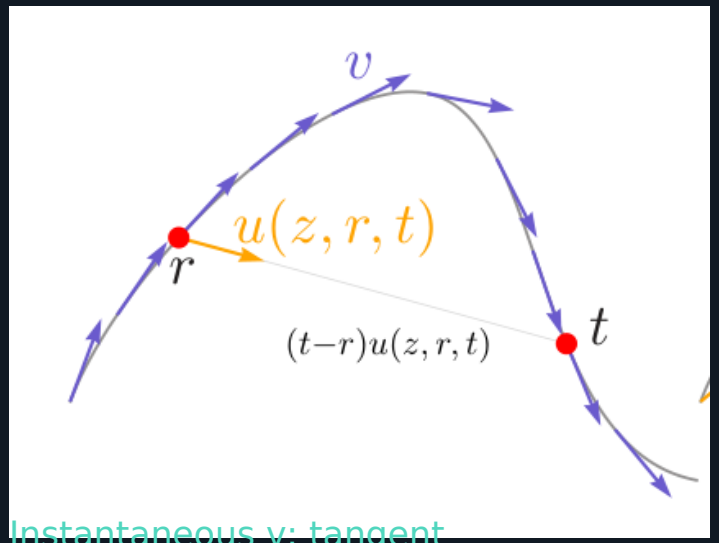
MeanFlow turns a calculus identity into a target.

Use h = t − r. Data are at t = 0; noise is at t = 1.

$$u(z_t, r, t) = \frac{1}{t-r}\int_r^t v_s(z_s)\,ds = \frac{z_t - z_r}{t-r}$$

$$v_t(z_t) = u(z_t, r, t) + (t-r)\frac{d}{dt}u(z_t, r, t)$$

$$u_{tgt} = \tilde{v} - (t-r)\,\mathrm{JVP}_{(z,r,t)}u_\theta\,[\tilde{v}, 0, 1]$$



Instantaneous v: tangent
Average u: displacement direction

Source: Fig. 3, left panel (cropped).

Train u against a stopped target; sample once: x̂ = ε − uθ(ε, 0, 1).

Generalize the normalization, keep the physical flow.

MeanFlow uses f(h) = h. A different f changes the learned representation.

$$h = t - r, \qquad f(h)w(z_t, r, t) = \int_r^t v(z_\tau, \tau)\, d\tau$$

$$f'(h)w + f(h)\frac{dw}{dt} = v, \qquad r \text{ fixed}$$

REGULARITY

Require f(0)=0 and f′(h)>0.
Keep a finite, nonzero local limit.

INFERENCE

Use displacement f(h)wθ directly.
No extra sampling evaluations.

Equivalent fields need not give identical updates.

With stop-gradient, the target is frozen during each parameter update.

$$w_{tgt} = \frac{\tilde{v} - f(h) D_\theta}{f'(h)}, \qquad D_\theta = \mathrm{JVP}(w_\theta; [\tilde{v}, 0, 1])$$

$$\mathcal{L}_f = \mathbb{E}\left[\lambda(h) \| w_\theta - \mathrm{sg}(w_{tgt}) \|_2^2\right]$$

Hypothesis: f changes target scaling and interval-dependent updates; some choices may improve training stability or sample quality.

Key alternative explanation: the benefit may be only interval reweighting.
A matched-weight control is essential; no improvement is claimed yet.

Use the paper’s CIFAR-10 benchmark and U-Net.

CIFAR-10: unconditional 32 × 32 RGB generation in pixel space; no VAE needed.

$$f_\beta(h) = h[1 + \beta h(1 - h)], \qquad \beta \in \{-0.5, 0, 0.5\}$$

MODEL & DATA

CIFAR-10 + ~55M U-Net.
Original f(h)=h baseline.
Same paper training recipe.

CANDIDATES

Fixed β = −0.5 / 0.5.
f′(0)=1 and f(1)=1.
Match interval weights.

MEASURE

FID-50K at 1 NFE.
3 seeds; matched budgets.
Gradient variation + time.

Reproduce the CIFAR-10 recipe; isolate reweighting in a squared-loss ablation.
Optional scale-up: ImageNet 256 × 256 with the paper’s DiT backbone.

Deliver a mechanism, not only a better metric.

Current status: paper and derivations checked; neural-model experiments are pending.

01 VERIFY

Analytic identity checked.
Matched-update check passed.
Next: automatic JVP tests.

02 IMPLEMENT

Reproduce the baseline.
Run candidates and controls.
Repeat with three seeds.

03 EXPLAIN

Separate scaling and weights.
Report negative results too.
Assess scope and prior work.

Research question: can a better normalization help beyond simple reweighting?